THE UNIVERSITY OF ZAMBIA

IN ASSOCIATION WITH

ZAMBIA ICT COLLEGE

JUNE 2022 - SESSIONAL EXAMINATIONS

OPERATING SYSTEMS CONCEPTS (SEC 2204)

INSTRUCTIONS TO CANDIDATES

1. Time allowed is **THREE (3) hours**
2. There are SIX **(6)** questions in this paper. Answer any **FIVE(5)**questions
3. Each question carries **TWENTY (20)** marks **[Total Marks = 100]**
4. You are required to answer any **FIVE (5)** questions
5. Neat and clearly written work is encouraged

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

QUESTION ONE

a) Define the following terminologies

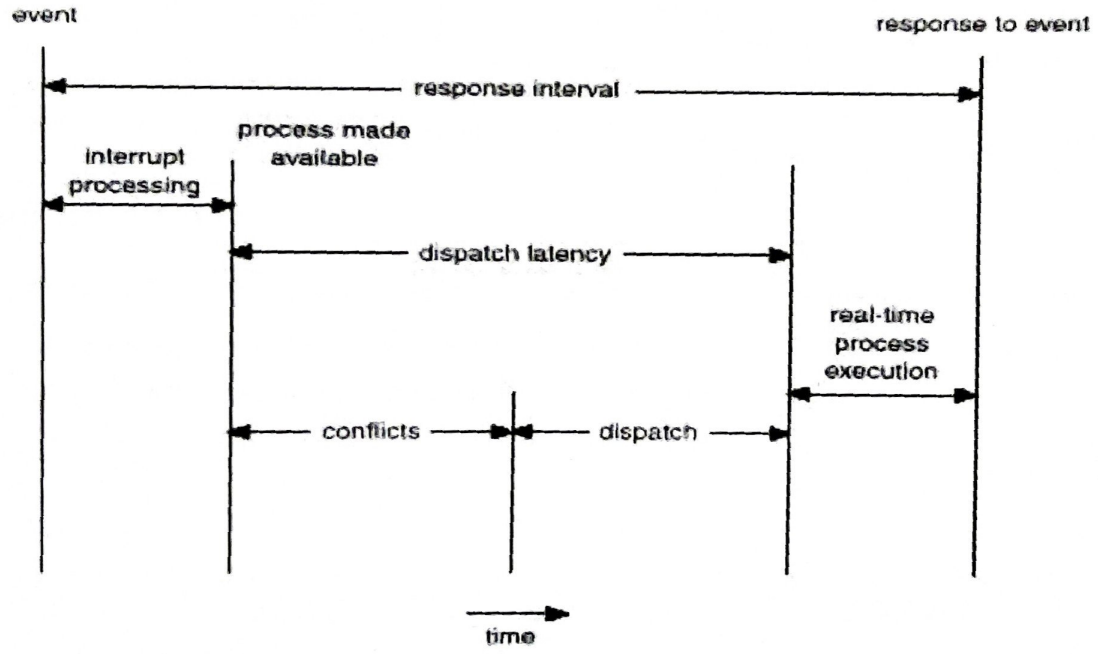
i. Deadlock [1 mark]
ii. Turnaround time [1 mark]
iii. Cache coherency [1 mark]
iv. Throughput [1 mark]
v. Timesharing [1 mark]
vi. Multiprogramming. [1 mark]

b) List three (3) condition required for a deadlock to be possible [3 Marks]

c) State three (3) advantage of Round-robin Scheduling [3 Marks]

d) Describe any three (3) important functions of an operating system [3 Marks]

e) Briefly describe each of the following terms shown in the diagram [5 Marks]



QUESTION TWO

a) Describe the differences between the following terms

i. Multiprocessing and Multiprogramming. [3 Marks]

ii. Multiprogramming and Timesharing **[3 Marks]**

iii. preemptive scheduling and non-preemptive scheduling. **[3 Marks]**

iv. logical addresses and physical addresses. **[3 Marks]**

b) Explain the difference between Job and Process? **[2 marks]**

c) Explain how Buffering can improve the performance of a Computer system? **[2 Marks]**

d) Assume the following processes, each with their arrival time and burst time. For SJF Non-Preemptive job scheduling, calculate the average waiting time. **[2 Marks]**

Process	Arrival time	Burst Time
P0	0 .0	7
P1	2.0	4
P2	4.0	1
P2	5.0	4

e) Sate two (2) drawbacks of using Round-robin scheduling **[2 marks]**

QUESTION THREE

a) Consider the following table of arrival time and burst time for three processes P0, P1 and P2.
The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

[2 Marks]

Process	Arrival time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

b) An operating system is a program on which application programs are executed and acts as a communication bridge (interface) between the user and the computer hardware. List and explain four (4) services the Operating System provide to the users. **[4 Marks]**

c) Explain one (1) benefit of using modules? **[1 Mark]**

d) Explain two (2) advantages of using a higher-level language to implement an operating system' **[2 Marks]**

e) Explain why an operating system may be viewed as a resource allocator of such things as CPU time, memory space, file-storage space, I/O devices **[2 Marks]**

f) Describe the following scheduling algorithms

i. Non-Pre-Emptive, First Come, First Serve **[3 Marks]**

ii. Round Robin **[3 Marks]**

iii. Shortest Job First **[3 Marks]**

QUESTION FOUR

(a) The following terms relate to how the operating system manages multiprogramming. Describe the part each plays in this process.

i. Process **[1 Mark]**

ii. Process management **[1 Mark]**

iii. Memory management **[1 Mark]**

iv. CPU scheduling **[1 Mark]**

(b) State three (3) memory management techniques and give the general approach taken in each. **[3 Marks]**

(c) The conceptual stages through which a process moves while being managed by the operating system. "*new, ready, running, waiting,* and *terminated*", Describe how a process might move through the various process states. Give a specific reason why this process moves from one state to another. **[3 Marks]**

(d) State and describe three (3) CPU scheduling algorithms. **[6 Marks]**

QUESTION FIVE

(a) State the number of instructions performed during 1 cycle on the CPU? **[3 Marks]**

(b) Describe the *three-state process model,* describe what transitions are valid between the three states, and describe an event that might cause such a transition. The three-state process model dictates that a process may take the form of one of three states, **[3 Marks]**

(c) Consider the following set of processes

Process	P1	P2	P3	P4	P5
Service time	120	60	180	50	300

i. Draw a Gantt chart that shows the completion times for each process using first-come, first served CPU scheduling. **[3 Marks]**

ii. Draw a Gantt chart that shows the completion times for each process using shortest-job-next CPU scheduling. **[3 Marks]**

(d) An operating system supplies system calls to allow you to COPY, DELETE and RENAME a file. Discuss the differences between using COPY/DELETE and RENAME to give a file new name? there is a performance impact in using copy/delete as the entire file is copied. If you use rename then only the index entry has to be changed. **[2 Marks]**

(e) Describe a sequence the sequence of step that occur when a timer interrupt occurs that eventually results in a context switch to another application. **[4 Marks]**

(f) Giving an appropriate example, explain a *race condition.* **[2 Marks]**

QUESTION SIX

(a) Define the following terminologies

i. Thrashing **[1 Mark]**

ii. Bus **[1 Mark]**

iii. Multitasking **[1 Mark]**

iv. Application Programming Interfaces (APIs). **[1 Mark]**

(b) Describe the difference between a 32 bit bus and a 64 bit bus **[2 Marks]**

(c) Describe five (5) critical characteristics of a bus **[3 Marks]**

(d) State the purpose of a swap (or page) file **[1 Marks]**

(e) List the correct sequence of actions that takes place during the boot-up process **[2 Marks]**

(f) There are two major design types for operating system kernels: monolithic kernels and microkernels. Which of these types better satisfies the following requirements? If both are equally as good write both. Justify your answers.

i. Convenient access to operating system data structures. **[2 Marks]**

ii. Addition of new operating system components. **[2 Marks]**

iii. Modification of operating system components. **[2 Marks]**

iv. Security and reliability. **[2 Marks]**

QUESTION SEVEN

(g) Define the term process and state any attributes of a process **[2 Marks]**

(h) Explain the relationship between a threads and a processes **[2 Marks]**

(i) What is the maximum file size supported by a file system with 16 direct blocks, single, double, and triple indirection? The block size is 512 bytes. Disk block numbers can be stored in 4 bytes. **[8 Marks]**

(j) Priorities make real-time scheduling more difficult. Give a reason why priorities are still used in real-time systems **[2 Marks]**

(k) Describe the two general roles of an operating system, and elaborate why these roles are important. **[2 Marks]**

(l) Multi-programming (or multi-tasking) enables more than a single process to apparently execute simultaneously. Explain how this achieved on a uniprocessor. **[4 Marks]**

THE END